MIL-C-22553(AS)
Amendment-3
28 March 1973
Superseding
Amendment-2
15 July 1963

MILITARY SPECIFICATION

CONVERTER, SIGNAL DATA CV-1053A/ARC

This amendment forms a part of Military Specification MIL-C-22553(AS) dated 1 November 1960 and has been approved by the Naval Air Systems Command, Department of the Navy.

By this amendment MIL-C-22553(WEP) shall be changed to MIL-C-22553(AS) and throughout the specification Bureau of Naval Weapons shall be changed to Naval Air Systems Command.

*Page 1, paragraph 1.2: Delete "CV-1053/ARC" and substitute "CV-1053A/ARC"

*Page 1, paragraph 2.1, under SPECIFICATIONS Military: Delete "MIL-E-4682" and SAR-300" and their titles and add

"MIL-M-7793	Meter, Time Totalizing
MIL-R-23094	Reliability Assurance for Production Acceptance of Avionic Equipment, General Specification for
MIL-S-23603	System Readiness/Maintainability; Avionics Systems Design, General Specification for"

Page 2, paragraph 2.1, under STANDARDS Military: Add

"MS 17321	Meter, Time Totalizing, 28V, DC, Minature Digital"

*Page 2, paragraph 3.1, first sentence: Add

"(See 6.4)"

FSC 5895

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*Page 2, paragraph 2.2, after "Commanding Officer": Delete and substitute

"Publications and Forms Center, Code 105,
5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120".

*Page 2, paragraphs 3.1.1, 3.1.2 and 3.1.2.1: Delete and substitute

"Nonstandard Parts and Material Approval - Approval for the use of nonstandard parts and materials (including transistors and diodes) shall be obtained as outlined in Specification MIL-E-5400. No vacuum tubes shall be employed. Only solid state electron devices shall be used".

Page 3, paragraph 3.2.3.1: Delete "500 hours" and substitute

"200 hours"

Page 3, paragraph 3.2.3.3: Delete and substitute

"3.2.3.3 Reliability in Mean Time Between Failures - The equipment shall have a minimum of 200 hours of mean (operating) time between failure when tested and accepted as outlined under the requirements of 4.4.2."

*Page 3: Add

"3.2.3.4 Time Totalizing Meter - A time totalizing meter conforming, as applicable, with specification MIL-M-7793 and MS 17321 shall be included in the CV-1053A/ARC."

Page 4, paragraph 3.3.2.1: Delete "±3 cycles" and substitute

"±6 cycles"

Page 5, paragraph 3.3.4, after third sentence: Add

"Under service conditions, the equipment shall operate satisfactorily with an input signal of from 0.2 to 10.0 Volts RMS."

Page 5, paragraph 3.3.7.1, first sentence: Delete and substitute

"A suitable switching means shall be provided which shall permit reversal of 'mark' and 'space' polarity for receiving only."

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Page 5, paragraph 3.3.9: Add

"Compliance is to be based upon random noise measured as RMS."

*Page 5, paragraph 3.4.1: Delete "CV-1053/ARC" and substitute "CV-1053A/ARC."

*Page 5, paragraph 3.4.1.2: Delete and substitute

"*3.4.1.2 Mark and Space Designation - A 2425 cycle per second tone shall represent "space" and be produced by connecting the audio frequency shift keyer input to the signal common line. A 1575 cycle per second tone representing "mark" shall be generated when the keyer input is open circuit."

Page 5, paragraph 3.4.1.4: Delete "3-7/8 inches" and substitute

"4-9/16 inches"

Page 5, paragraph 3.4.1.6, second sentence: Delete and substitute

"The circuitry of each should be independent except for the two band pass filters which shall be highly reliable, hermetically sealed components. Malfunction or damage to one function will not impair operation of the other."

*Page 6, paragraph 3.4.2: Delete and substitute

"3.4.2 Mounting MT-2427/ARC - The mounting base for the signal data converter shall be capable of installation in any attitude. A multifunctional barrier vibration isolator of a flexible plastic type shall be employed subject to approval of the procuring activity. Means shall be provided to mount or remove the Signal Data Converter from the mounting base without special tools. The weight of the mounting shall not exceed 0.3 pound."

Page 8, paragraph 4.4.(2): Delete and substitute

"(2) Reliability Quality Assurance Tests"

Page 9, paragraphs 4.4.2 and 4.4.2.1: Delete and substitute

"4.4.2 Reliability Quality Assurance Tests - Equipments which have successfully passed the individual tests shall be tested for reliability assurance in accordance with specification MIL-R-23094, Procedure I.

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4.4.2.1 Test Level - The Test Level shall be Level III from Table I of specification MIL-R-23094 except that vibration testing shall be limited to 20 cps and the length of the heating cycle shall be based on a stabilized high temperature of two hours. The duty cycle shall not exceed four hours.

4.4.2.2 Accept-Reject Criteria - Figure 12 of specification MIL-R-23094 and an MTBF of 200 hours shall be used to make accept/reject decisions.

4.4.2.3 Performance Characteristics to be Measured - (Measurement to be made at end of duty cycle)

Characteristic	Specification Requirement
Frequency Characteristic	3.3.2
Sensitivity	3.3.4
Selectivity	3.3.5

4.4.2.4 Quantity to be Tested -

4.4.2.4.1 Qualification Plan - The minimum quantity of production equipment to be tested for the qualification plan shall be in accordance with 3.2.3.1 of specification MIL-R-23094.

4.4.2.4.2 Sample Plan - The minimum quantity of production equipment to be tested for the sample plan shall be in accordance with 3.2.3.2 of specification MIL-R-23094; however, the largest possible sample quantity is recommended to reduce total test time.

4.4.2.5 Determination of Compliance - Compliance shall be determined by the accept-reject criteria of specification MIL-R-23094, Figure 12.

4.4.2.6 Failure Criteria - In addition to the general requirements of specification MIL-R-23094, whenever performance requirements fall below those of 4.4.2.3 at least one failure has occurred."

*Page 12, paragraph 4.9: Delete

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Page 13, paragraph 6.2: Add

"(4) Reordered Equipment - The following specifications shall take precedence over those concerning performance requirements listed or referenced in 2.1, where equipment is reordered.

MIL-E-5400E	26 June 1961
MIL-T-5422E, Amendment-2	15 November 1961
MIL-I-6181D	25 November 1959
MIL-R-23094A	15 January 1963"

Page 13, paragraph 6.4: Add

"Such information shall include consideration of System Readiness/Maintainability requirements of specification MIL-S-23603."

Project Number 5895-N224

Note: An asterisk preceding a page number denotes a change to the specification which did not appear in the previous amendment.

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